Introducción a la Dinámica de Fluidos Computacional y Fenómenos de Transporte Computacional

INFORME TÉCNICO - EXAMEN FINAL

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1 Introduction

The governing equations of fluid mechanics form a nonlinear, coupled system whose solution gives the primitive variables of a flow: velocity, pressure, density, and temperature. Because the full system is analytically intractable (except in simple situations), it is useful to study a single scalar equation that reproduces the essential physical mechanisms of the full equations while remaining tractable. Such a model equation must contain, at minimum, an unsteady term, a nonlinear convective term, and a diffusive term.

The Burgers equation [1] provides exactly such a model. In its complete form it is written

$$\underbrace{\frac{\partial u}{\partial t}}_{\text{transient}} + \underbrace{u\,\frac{\partial u}{\partial x}}_{\text{convective}} = \underbrace{\mu\,\frac{\partial^2 u}{\partial x^2}}_{\text{viscous}}, \tag{1}$$

where the three terms represent, respectively, the temporal evolution of the field, the nonlinear convection of the field by itself, and the viscous (diffusive) transport governed by the coefficient μ. Retaining the viscous term, the equation is parabolic. When that term is omitted, the remaining equation is hyperbolic, and it is called the inviscid Burgers equation:

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} = 0. \tag{2}$$

This inviscid form may be regarded as a one-dimensional analogue of the Euler equations for an inviscid flow. Its main difference from the linear convection equation ($u_t + c\,u_x = 0$, where $c =$

constant) is that the convective wave speed equals the local value of the transported variable u, so that different points of the solution propagate at different speeds. This variable wave speed causes the convergence of characteristics and the consequent formation of discontinuous solutions, analogous to the shock waves of compressible gas dynamics. The inviscid Burgers equation is therefore a simple but effective model for studying the formation and propagation of shocks in one dimension.

1.1 Mathematical modelling

The inviscid Burgers equation is not introduced arbitrarily, but it can be derived directly from the conservation of momentum for an inviscid, one-dimensional flow [1]. The starting point is the momentum equation in conservative form,

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho\, \mathbf{u}\, u) = \nabla \cdot \boldsymbol{\tau}_{ij} - \nabla p + \rho\, \mathbf{g} + S_u, \tag{3}$$

in which the left-hand side collects the temporal variation and the convective transport of momentum, and the right-hand side gathers the viscous stresses, the pressure gradient, the gravitational body force, and any additional momentum source.

Two simplifications are introduced. First, the fluid is assumed inviscid ($\mu = 0$), so the viscous stress term vanishes. Second, no body forces or additional momentum sources are considered, so the gravitational and source terms are discarded. The pressure gradient is likewise omitted, leaving the balance between the unsteady and convective terms alone. Equation (3) then reduces to

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho\, \mathbf{u}\, u) = 0. \tag{4}$$

Restricting the problem to a single spatial dimension, the divergence reduces to a derivative with respect to x, and the equation becomes

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} = 0. \tag{5}$$

The two conservative terms are now expanded by the product rule. Applying it to the temporal term gives $\partial_t(\rho u) = \rho\, \partial_t u + u\, \partial_t \rho$, so that

$$\rho\, \frac{\partial u}{\partial t} + u\, \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u u)}{\partial x} = 0. \tag{6}$$

Expanding the convective term in the same manner, $\partial_x(\rho u \cdot u) = u\, \partial_x(\rho u) + \rho u\, \partial_x u$, yields

$$\rho\, \frac{\partial u}{\partial t} + u\, \frac{\partial \rho}{\partial t} + u\, \frac{\partial(\rho u)}{\partial x} + \rho u\, \frac{\partial u}{\partial x} = 0. \tag{7}$$

The second and third terms share the common factor u and can be grouped together. The quantity inside the resulting bracket is precisely the mass-conservation (continuity) equation,

$$\rho\, \frac{\partial u}{\partial t} + u\, \underbrace{\left[\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x}\right]}_{\text{continuity} = 0} + \rho u\, \frac{\partial u}{\partial x} = 0, \tag{8}$$

which vanishes under the continuity assumption. After the cancellation, the equation reads

$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} = 0. \tag{9}$$

Dividing by the density ρ removes it from both terms and leaves the non-conservative form of the inviscid Burgers equation,

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0. \tag{10}$$

For the numerical treatment that follows, the conservative (flux) form is required, since only this form reproduces the correct propagation speed of discontinuous solutions. The two forms are connected by the chain rule, which gives $u\,\partial_x u = \partial_x(u^2/2)$. Substituting this identity into equation (10) produces the conservative form, which may be written compactly by defining the flux function $F = u^2/2$:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \iff \frac{\partial u}{\partial t} + \frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) = 0 \iff \frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0, \qquad F = \frac{u^2}{2}. \tag{11}$$

The non-conservative and conservative forms are equivalent for smooth solutions, where the chain rule holds. They differ, however, at discontinuities, where the derivative $u\,\partial_x u$ is undefined; there, only the flux form retains physical meaning.

1.2 PDE characterization

The inviscid Burgers equation (11) is a first-order, nonlinear, hyperbolic partial differential equation. The hyperbolic nature of the equation means that the solution propagates at finite speed along well-defined paths in the (x, t) plane, called *characteristics* [2].

Characteristics. A characteristic is a curve in the (x, t) plane along which the solution remains constant. Consider how u varies along a path $x(t)$. By the multivariable chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t} u(x(t), t) = \frac{\partial u}{\partial t} + \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\partial u}{\partial x}. \tag{12}$$

If the path is chosen so that its slope equals the local wave speed,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = u, \tag{13}$$

the right-hand side of (12) becomes $u_t + u\,u_x$, which is zero by the equation itself. Therefore u is constant along that path. Since u is constant, the slope $\mathrm{d}x/\mathrm{d}t = u$ is also constant, and each characteristic is a *straight line* whose slope is set by the value it carries. Each point of the initial profile then travels at a speed equal to its own value, transporting that value unchanged.

This is the essential feature that distinguishes the Burgers equation from the linear convection equation: in the linear case all characteristics share the same speed c and never cross, while here faster points overtake slower ones, so the characteristics may intersect each other.

Shock formation. Where a faster characteristic catches a slower one ahead of it, the two cross. At that instant the profile becomes vertical and a discontinuity, a *shock*, appears. The time of the first crossing follows directly from the geometry of the characteristics. A characteristic starting at ξ carries the value $u_0(\xi)$ and is the straight line

$$x(\xi, t) = \xi + u_0(\xi)\, t. \tag{14}$$

Two neighbouring characteristics, starting at ξ and $\xi + \mathrm{d}\xi$, cross when their separation closes to zero. Differentiating (14) with respect to the starting point gives that separation,

$$\frac{\partial x}{\partial \xi} = 1 + u_0'(\xi)\, t, \tag{15}$$

which vanishes when

$$t = \frac{-1}{u_0'(\xi)}. \tag{16}$$

This is positive only where $u_0'(\xi) < 0$, that is, on the descending (forward) face of the profile, where neighbouring characteristics converge. The first crossing, the shock, occurs at the smallest time t, which corresponds to the most strongly negative slope of the initial data. Taking the minimum over ξ gives the *breaking time*

$$t_b = \frac{-1}{\min\limits_{\xi} u_0'(\xi)}. \tag{17}$$

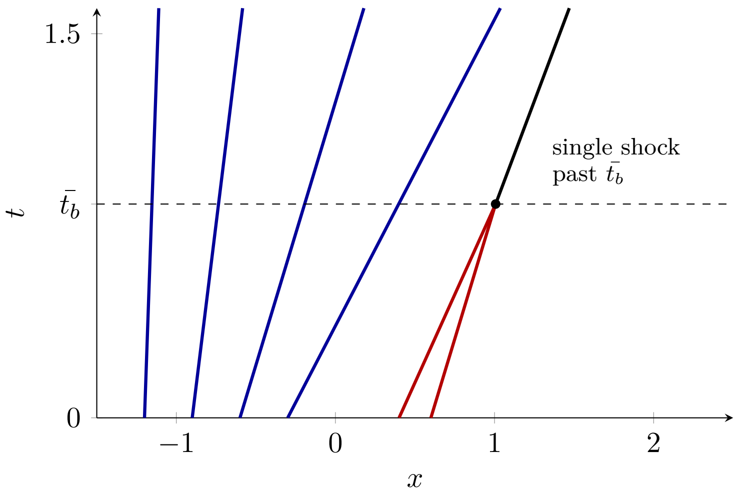


Figure 1: Characteristics of Configuration 1 in the (x, t) plane, each carrying its initial value at constant speed along the straight line $x = \xi + u_0(\xi)\, t$. The rear characteristics (blue) diverge, producing a rarefaction, with the slower points (smaller u_0) tracing steeper lines. The two representative forward characteristics (red) converge and first meet at $\bar{t_b} > t_b \approx 0.824$; beyond this point they no longer cross but merge into a single shock (black), which propagates at the Rankine–Hugoniot speed.

Conservative form and the Rankine–Hugoniot condition. Once a discontinuity is present, the derivative $u\, u_x$ in the non-conservative form (10) is undefined, and the two forms of the equation are no longer equivalent. Only the conservative (flux) form remains meaningful across a discontinuity, because it expresses a balance of fluxes rather than of derivatives.

The speed of the discontinuity follows from this balance. Let the shock sit at position $x_s(t)$, separating a left state u_L from a right state u_R, and integrate the conservative form over a fixed interval $[x_1, x_2]$ that contains it:

$$\int_{x_1}^{x_2} \frac{\partial u}{\partial t}\, \mathrm{d}x + \int_{x_1}^{x_2} \frac{\partial F}{\partial x}\, \mathrm{d}x = 0. \tag{18}$$

Since the interval is fixed in time, the time derivative may be taken outside the first integral; the second integral is evaluated by the fundamental theorem of calculus, giving the flux at the two endpoints:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{x_1}^{x_2} u\,\mathrm{d}x + \left[\,F(u_2) - F(u_1)\,\right] = 0, \tag{19}$$

where $u_1 = u(x_1)$ and $u_2 = u(x_2)$. Rearranging,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{x_1}^{x_2} u\,\mathrm{d}x = F(u_1) - F(u_2), \tag{20}$$

which states that the amount of u stored in the interval changes only through the fluxes entering at x_1 and leaving at x_2. Splitting the integral at the shock and taking the interval to shrink onto the shock ($x_1 \to x_s^-$ and $x_2 \to x_s^+$), so that u_L and u_R denote the limits of u immediately behind and ahead of the discontinuity, the stored quantity over the thin interval is $u_L(x_s - x_1) + u_R(x_2 - x_s)$. Differentiating in time, only the moving limit x_s contributes,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{x_1}^{x_2} u\,\mathrm{d}x = (u_L - u_R)\,\frac{\mathrm{d}x_s}{\mathrm{d}t}. \tag{21}$$

Taking x_1, x_2 just outside the shock, so that $F(u_1) = F(u_L)$ and $F(u_2) = F(u_R)$, and equating (20) and (21) gives the shock speed $s = \mathrm{d}x_s/\mathrm{d}t$,

$$s = \frac{F(u_L) - F(u_R)}{u_L - u_R}, \tag{22}$$

the *Rankine–Hugoniot condition* [2]. For the Burgers flux $F = u^2/2$ it reduces to the average of the two states,

$$s = \frac{\frac{1}{2}(u_L^2 - u_R^2)}{u_L - u_R} = \frac{u_L + u_R}{2}. \tag{23}$$

The shock therefore travels at the mean of the velocities it separates.

1.3 Configurations

The inviscid Burgers equation (11) is solved on a bounded domain $-L \leq x \leq L$ for $t \geq 0$, subject to Dirichlet boundary conditions $u(-L, t) = u_L$ and $u(L, t) = u_R$ and to a prescribed initial condition $u(x, 0) = u_0(x)$. Two configurations are considered: they differ in the shape of the initial profile, in the size of the domain, and in the boundary values.

Configuration 1: Gaussian profile. The first configuration takes a smooth Gaussian profile as initial condition,

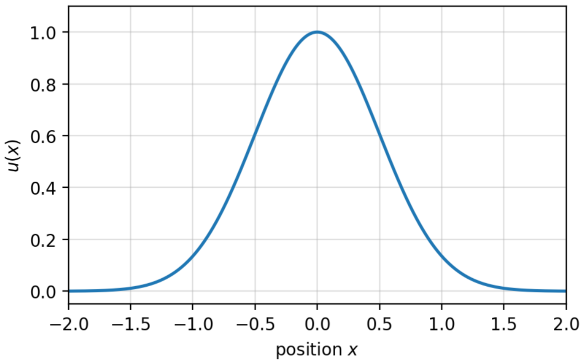
$$u_0(x) = e^{-2x^2}, \tag{24}$$

on the domain $L = 2$, with boundary values $u_L = u_R = 0$. The profile is symmetric about $x = 0$, with a maximum of 1 and a minimum of 0, as shown in Figure 2a. Because the field is everywhere non-negative, the wave speed u is non-negative throughout the whole domain, and the pulse propagates towards positive x.

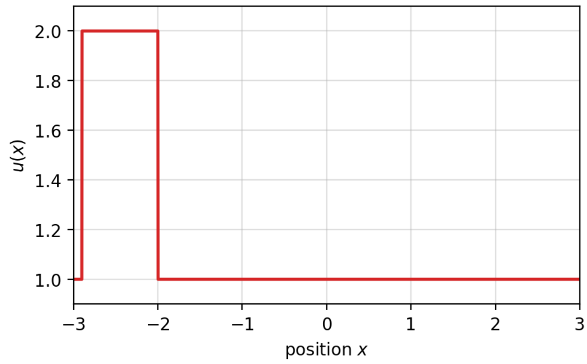
Configuration 2: Square wave. The second configuration takes a discontinuous, square-wave profile,

$$u_0(x) = \begin{cases} 2, & -2.9 \leq x \leq -2, \\ 1, & \text{elsewhere}, \end{cases} \tag{25}$$

on the domain $L = 3$, with boundary values $u_L = u_R = 1$. The profile consists of a band of elevated velocity $u = 2$ and width 0.9, located over the interval $[-2.9, -2]$, standing over a background where $u = 1$, as shown in Figure 2b.



(a) Configuration 1: Gaussian, $u_0(x) = e^{-2x^2}$ on $L = 2$.



(b) Configuration 2: Square wave, $u_0 = 2$ on $[-2.9, -2]$ and $u_0 = 1$ elsewhere, on $L = 3$.

Figure 2: Initial conditions for the two configurations under study.

In both configurations the velocity field is non-negative, so the convective transport is directed towards positive x and the upwind value at each face is taken from the control volume immediately to its west.

The two configurations are chosen to test the equation against qualitatively different initial data. The Gaussian profile is smooth and infinitely differentiable everywhere (C^∞), so it tests whether the scheme reproduces the spontaneous steepening of an initially regular field. The square wave is discontinuous from the beginning, so it shows the treatment of sharp fronts already present in the data. Together they cover two interesting situations for a nonlinear hyperbolic equation: the emergence of a discontinuity from smooth data and the transport of a discontinuity that exists from the start.

2 Development

2.1 Mesh

The spatial domain $[-L, L]$ is divided into N control volumes (cells) of uniform width

$$\Delta x = \frac{2L}{N}. \tag{26}$$

A cell-centred arrangement is adopted: the discrete values of the field are stored at the centres of the cells, while the cell faces are laid at a $\frac{\Delta x}{2}$ distance from each centre. The centre of the i-th interior cell is located at

$$x_i = -L + \left(i - \tfrac{1}{2}\right) \Delta x, \qquad i = 1, 2, \ldots, N, \tag{27}$$

so that the first and last centres are placed half a cell width inside the domain, at $x_1 = -L + \Delta x/2$ and $x_N = L - \Delta x/2$. The physical boundaries $x = \pm L$ therefore fall on cell faces rather than on stored nodes.

Each interior cell P is bounded by a western face w and an eastern face e, shared respectively with its neighbours W and E. The field is stored at the cell centres, while the convective fluxes are evaluated at the faces.

The Dirichlet boundary conditions are imposed by two ghost cells, one beyond each end of the domain. The western ghost cell holds the value u_L and the eastern ghost cell holds the value u_R; these supply the boundary-face fluxes required to update the first and last interior cells. Because in the two configurations the flow is everywhere directed towards positive x, the right boundary is an outflow boundary. With first-order upwind differencing, the update of the last interior cell does not depend on the downstream ghost value, so the prescribed right-boundary value does not influence the solution; the shock convects out of the domain without reflection. The left (inflow) boundary value, by contrast, propagates into the domain and is then imposed.

The full array of stored values therefore contains $N+2$ values: the N interior cells that are advanced by the numerical scheme, and the two ghost cells that carry the prescribed boundary information. The number of cells N is a parameter of the discretization. A coarse mesh resolves the solution only approximately, while a fine mesh represents sharp gradients more faithfully at the cost of a smaller time step and a larger number of iterations.

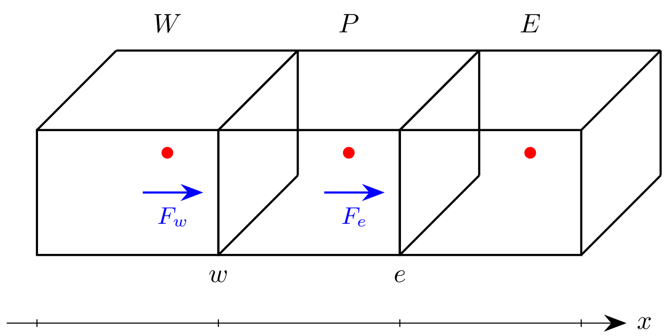


Figure 3: Cell-centred finite-volume mesh: control volumes with stored centres (W, P, E) and convective fluxes evaluated at the faces (w, e).

2.2 Discretization

The conservative form of the inviscid Burgers equation (11) is discretized by the finite-volume method [1]. The starting point is the integration of the equation over a control volume and over a time step $[\,t,\, t + \Delta t\,]$,

$$\int_t^{t+\Delta t}\int_{CV} \frac{\partial u}{\partial t}\, dV\, dt + \int_t^{t+\Delta t}\int_{CV} \frac{\partial F}{\partial x}\, dV\, dt = 0, \tag{28}$$

where the first term is the unsteady contribution and the second is the convective contribution, with flux $F = u^2/2$.

In the unsteady term the order of integration is exchanged, so that the time integral is evaluated first. This exchange is permitted by Fubini's theorem, since the control volume is fixed in time and the integrand is bounded and integrable over the space–time domain:

$$\int_{CV}\int_t^{t+\Delta t} \frac{\partial u}{\partial t}\, dt\, dV + \int_t^{t+\Delta t}\int_{CV} \frac{\partial F}{\partial x}\, dV\, dt = 0. \tag{29}$$

The convective term is transformed by the Gauss (divergence) theorem, which replaces the volume integral of the flux derivative by the integral of the flux over the boundary ∂V of the control volume. The inner time integral of the unsteady term is evaluated by the fundamental theorem of calculus, giving the difference between the value of u at the new and old time levels:

$$\int_{CV} \left[u_P - u_P^0 \right] dV + \int_t^{t+\Delta t}\int_{\partial V} F\, dS\, dt = 0, \tag{30}$$

where the superscript 0 denotes the known value at the start of the time step, while the subscript P denotes the value of u at cell P.

For the one-dimensional control volume of width Δx and cross-sectional area A, the remaining volume integral of the unsteady term reduces to $(u_P - u_P^0)\, \Delta x\, A$, and the surface integral reduces to the difference between the fluxes through the eastern and western faces, F_e and F_w:

$$\left(u_P - u_P^0 \right) \Delta x\, A + \int_t^{t+\Delta t} (F_e - F_w)\, A\, dt = 0. \tag{31}$$

The cross-sectional area A appears as a common factor in both terms and is cancelled. The remaining time integral of the face fluxes is then approximated. Because the flux $F = u^2/2$ is nonlinear, an explicit (forward in time) scheme is chosen, so that the fluxes are evaluated at the old time level; the integral over the step is approximated by the old-level integrand multiplied by Δt:

$$\left(u_P - u_P^0 \right) \Delta x + \left(F_e^0 - F_w^0 \right) \Delta t = 0. \tag{32}$$

Solving for the new value u_P gives the explicit update used to advance the solution:

$$u_P = u_P^0 - \frac{\Delta t}{\Delta x} \left(F_e^0 - F_w^0 \right). \tag{33}$$

Each cell is advanced directly from the known values of the previous time level, with no coupled system to solve. The choice of the time step Δt, restricted by stability, and the evaluation of the face fluxes F_e^0 and F_w^0 by the upwind scheme are addressed in the following subsections.

2.3 Upwind differencing scheme (UDS)

The explicit update (33) requires the convective fluxes at the cell faces, F_e^0 and F_w^0, but the field is stored at the cell centres. A scheme is therefore needed to express the face values in terms of the centre values. Because the problem contains convection but no diffusion, a central or hybrid scheme (CDS, HDS) is not appropriate: such schemes interpolate symmetrically between neighbouring nodes and disregard the direction in which information physically propagates. The upwind differencing scheme (UDS) is used instead, in which the face value is taken directly from the neighbouring cell that lies on the upwind side of the face [3]. For a generic transported quantity ϕ and the eastern face e,

$$\phi_e = \begin{cases} \phi_P, & F_e \geq 0, \\ \phi_E, & F_e < 0, \end{cases} \tag{34}$$

where F_e is the convective flux through face e. The face value is thus determined exclusively by the direction of the flow at that face.

In the present problem the transported quantity is the velocity itself, $\phi = u$, and the upwind direction at each face is set by the sign of the local wave speed. Since the Burgers equation is nonlinear in u, the convective flux cannot be used as the upwind indicator as in equation (34): the flux $F = u^2/2$ is always non-negative and therefore carries no information on the direction of propagation. The relevant wave speed is instead $f'(u) = u$, the velocity itself. The face velocity is estimated from the cell-centre value, $v = u_P$; its sign captures the direction of propagation, so that $v > 0$ corresponds to convective transport towards positive x. In both configurations studied this holds throughout the domain. The upwind rule applied to the eastern and western faces of cell P then gives

$$u_e = \begin{cases} u_P, & v \geq 0, \\ u_E, & v < 0, \end{cases} \qquad u_w = \begin{cases} u_W, & v \geq 0, \\ u_P, & v < 0. \end{cases} \tag{35}$$

For a positive face velocity the donor cell lies to the west, and for a negative face velocity it lies to the east; the scheme therefore adapts automatically to the direction of propagation. In both configurations studied here the velocity is non-negative throughout, so the western branch is always selected and the face fluxes are taken from the upwind (western) cell,

$$u_e = u_P, \qquad u_w = u_W. \tag{36}$$

Substituting the flux definition $F = u^2/2$, evaluated at the upwind face values and at the old time level, into the explicit update (33) gives the final discrete equation implemented in the solver:

$$u_P = u_P^0 - \frac{\Delta t}{\Delta x}\left[\frac{\left(u_P^0\right)^2}{2} - \frac{\left(u_W^0\right)^2}{2}\right]. \tag{37}$$

2.4 Boundary conditions

Both configurations are subject to Dirichlet boundary conditions, in which the value of the solution is prescribed at each end of the domain,

$$u(-L, t) = u_L, \qquad u(L, t) = u_R. \tag{38}$$

Because the mesh is cell-centred, the physical boundaries $x = \pm L$ fall on cell faces rather than on stored nodes. The boundary conditions are therefore imposed by means of two ghost cells, one placed beyond each end of the domain. The western ghost cell holds the value u_L and the eastern ghost cell holds the value u_R. These ghost values supply the boundary-face fluxes required to update the first and last interior cells, so that the prescribed boundary information enters the solution through the same flux-balance mechanism used in the interior. However the ghost cells are not advanced by the scheme, their values are kept fixed during the whole run.

For a hyperbolic equation the appropriate treatment of a boundary depends on the direction in which information propagates there, since information travels along the characteristics at the local wave speed u. At an *inflow* boundary the characteristics enter the domain, so the solution there is determined from outside; the boundary value must be prescribed, and a Dirichlet condition is the correct treatment. At an *outflow* boundary the characteristics leave the domain, so the solution there is determined by the interior; in principle the value should not be prescribed independently, but should follow from the solution as it convects out.

In both configurations the velocity is non-negative throughout the domain, so the convective transport is everywhere directed towards positive x. The left boundary is therefore an inflow boundary and the right boundary an outflow boundary. At the left (inflow) boundary the prescribed value u_L is imposed, which is both the correct treatment and the value required by the problem statement. The right (outflow) boundary requires more care, since imposing a fixed Dirichlet value there is, in general, not the physically correct treatment for an outflow boundary. In the present situation, however, the fixed value u_R is harmless because the upwind scheme never reads the eastern ghost cell.

For the rightward flow considered here, the first-order upwind discretization evaluates the flux of the last interior cell from its upwind (western) side. The eastern ghost value is therefore never selected by the scheme, and the solution at the outflow boundary is imposed entirely by the interior. The prescribed value u_R does not enter the computation, so its specific value cannot influence the result.

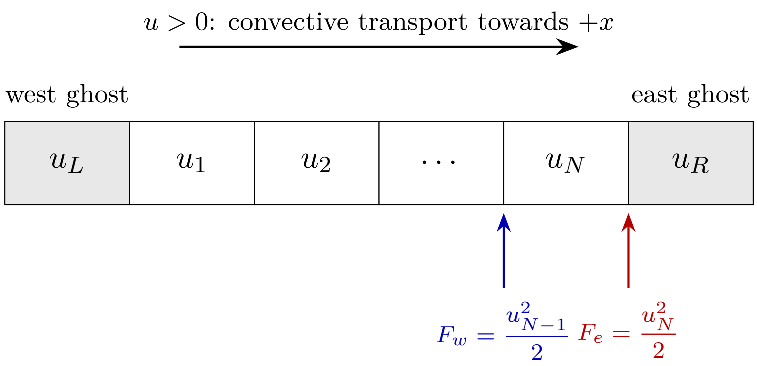


Figure 4: Outflow boundary treatment. For the rightward flow ($u > 0$) the first-order upwind scheme evaluates the fluxes of the last interior cell u_N from its upwind (western) side, so the eastern ghost value u_R is never read.

This mechanism is what makes the outflow treatment correct: in Configuration 1 the Gaussian pulse does reach the right boundary before the final time, so the boundary is genuinely disturbed. Because the outflow flux is taken from the interior side, the pulse is convected out cleanly and the fixed value u_R plays no role. In Configuration 2 the wave remains within the domain at all reported times, so the right boundary stays at its undisturbed background value $u = 1$; here the fixed Dirichlet value coincides with the solution and is likewise harmless. In both cases, therefore, the right boundary is correctly handled, for the single reason that the upwind scheme draws the outflow flux from the interior and never from the prescribed eastern ghost cell.

2.5 Timestep and the CFL condition

The explicit scheme is only conditionally stable: the time step cannot be chosen freely, but must satisfy a stability limit. This limit is established by examining the linear convection equation, which is similar to the Burgers equation, but with the variable wave speed u replaced by a constant speed c,

$$\frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = 0. \tag{39}$$

Discretizing equation (39) with the same explicit, first-order upwind scheme used for Burgers (for $c > 0$, the upwind neighbour lies to the west) gives

$$\frac{u_P - u_P^0}{\Delta t} + c\,\frac{u_P^0 - u_W^0}{\Delta x} = 0. \tag{40}$$

Solving for the new value u_P and collecting terms,

$$u_P = u_P^0 \left(1 - c\,\frac{\Delta t}{\Delta x}\right) + c\,\frac{\Delta t}{\Delta x}\,u_W^0. \tag{41}$$

Introducing the Courant number,

$$\mathrm{CFL} = c\,\frac{\Delta t}{\Delta x}, \tag{42}$$

the update takes the compact form

$$u_P = (1 - \mathrm{CFL})\,u_P^0 + \mathrm{CFL}\,u_W^0. \tag{43}$$

Equation (43) expresses the new value as a weighted combination of the old value at P and the old value at its upwind neighbour W. For the scheme to be stable, this combination must be *convex*: both weights must be non-negative, so that the updated value remains bounded by the values from which it is formed and cannot grow without limit. The weight on u_W^0 is CFL, which is non-negative, while the weight on u_P^0 is $1 - \mathrm{CFL}$, which is non-negative only if

$$|\mathrm{CFL}| = \left| c\,\frac{\Delta t}{\Delta x} \right| \le 1. \tag{44}$$

If this condition is violated, the weight $1 - \mathrm{CFL}$ becomes negative, the update ceases to be a convex combination, and small perturbations are amplified at each step, rendering the scheme unstable. Condition (44) is the Courant–Friedrichs–Lewy (CFL) condition [3].

For the Burgers equation the wave speed is not a constant c but the local value of the solution, u. Since the scheme is explicit, the speed is evaluated at the known (old) time level, so the local Courant number of cell P is $u_P^0\,\Delta t/\Delta x$. Stability requires the condition to hold at *every* cell simultaneously, so the constraint is set by the cell of largest velocity. Imposing $|\mathrm{CFL}| \le 1$ for all cells and solving for the time step gives

$$\Delta t \;\le\; \frac{\Delta x}{\max\limits_P \left|u_P^0\right|}. \tag{45}$$

Because the maximum velocity varies as the solution evolves, the limiting time step is recomputed at the beginning of each step from the current field, using a prescribed Courant number C:

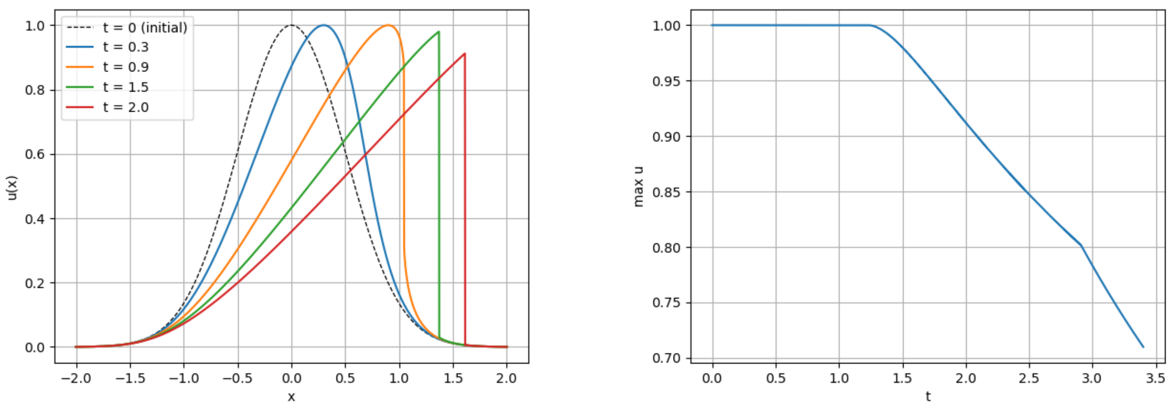
$$\Delta t = C \, \frac{\Delta x}{\max\limits_{P} \left| u_P^0 \right|}, \qquad C = 1. \tag{46}$$

The value $C = 1$ sits exactly at the stability limit and is admissible here. The first-order upwind update is a convex combination of old-time values: from equation (43), u_P is a weighted average of u_P^0 and u_W^0 with non-negative weights summing to one whenever CFL ≤ 1. A convex combination cannot exceed the largest of its arguments, so the new value at each cell is bounded by the neighbouring old values and $\max_P |u_P|$ cannot increase from one step to the next. The maximum velocity therefore does not grow in time, and the speed used to size the step at the start of each step also bounds the speed throughout the step. In addition, the time step is reduced where necessary so that the solution lands exactly on the prescribed output times $t_1, \ldots, t_4$ and on the final time t_f.

2.6 Results

The solver was run for both configurations on a uniform mesh of $n_{\text{vol}} = 5000$ control volumes up to the final time $t_f = 3.4$, with the explicit upwind scheme and adaptive time step at Courant number $C = 1$. The solution was recorded at the prescribed output times $t_1 = 0.3$, $t_2 = 0.9$, $t_3 = 1.5$ and $t_4 = 2.0$.

Configuration 1: Gaussian profile. The initial Gaussian pulse (24), of unitary peak on the background $u = 0$, is convected to the right at a local speed equal to its own amplitude. Since the higher portions of the pulse travel faster than the lower ones, the forward face steepens progressively into a shock while the trailing face spreads into a rarefaction, as shown in Figure 5a. The peak is transported downstream and its amplitude decays once the shock has formed, as plotted in Figure 5b. This monotonic decay is consistent with the convex combination principle established in Section 2.5.



(a) Solution $u(x)$ at $t = 0$, 0.3, 0.9, 1.5, 2.0.

(b) Evolution of $\max_x u(x, t)$.

Figure 5: Configuration 1: (a) solution profiles at the output times, showing the forward face steepening into a shock; (b) maximum value over time, which stays at $u = 1$ until the shock erodes the peak, then decays.

At the monitoring location x_A the value remains close to the background while the pulse is still upstream, then rises sharply as the shock front passes and decays once the peak has moved beyond it. The instant at which $u(x_A)$ departs from the background marks the arrival of the front at x_A and depends on the chosen position.

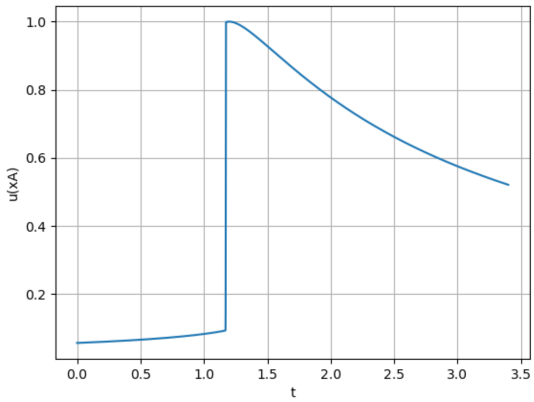
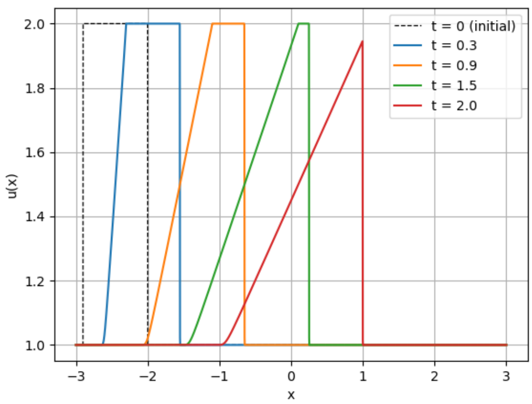


Figure 6: Configuration 1: time history of the solution at the probe location $x_A = 1.2$. The value rises gradually from the small initial Gaussian tail as the pulse is convected past x_A, reaches its maximum when the peak crosses the probe, and then decays as the shock front moves downstream beyond it.

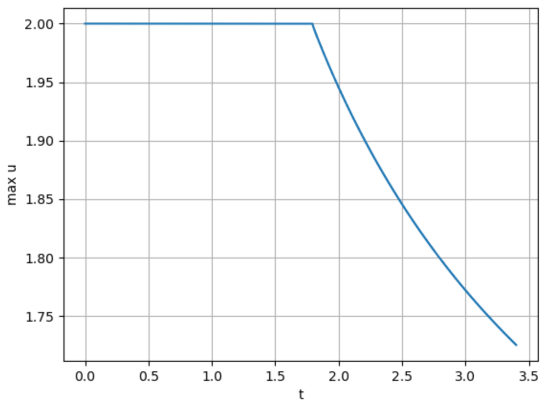
Configuration 2: Square wave. The initial condition is a block of amplitude $u = 2$ on $x \in [-2.9, -2]$ standing on a uniform background $u = 1$. The leading edge, separating the states $u_L = 2$ and $u_R = 1$, propagates as a shock at the Rankine–Hugoniot speed

$$s = \frac{F(2) - F(1)}{2 - 1} = \frac{2 - \frac{1}{2}}{1} = \frac{3}{2}, \tag{47}$$

while the trailing edge spreads into a rarefaction fan. The peak amplitude stays at $u_{\max} = 2.0$ until the rarefaction reaches the top of the block, after which it starts to decrease, as per Figure 7b.



(a) Solution $u(x)$ at $t = 0$, 0.3, 0.9, 1.5, 2.0.



(b) Evolution of $\max_x u(x, t)$.

Figure 7: Configuration 2: (a) solution profiles at the output times, showing the leading shock advancing at $s = 3/2$ and the trailing rarefaction fan; (b) maximum value over time, which stays at $u = 2$ until the rarefaction erodes the plateau and then decays.

At the fixed location x_A the value remains at the background until the shock front arrives, at which point it jumps up to the value immediately behind the shock. As the trailing rarefaction subsequently reaches x_A, the value decays back towards the background. The instant of the jump marks the passage of the front at x_A and depends on the chosen position.

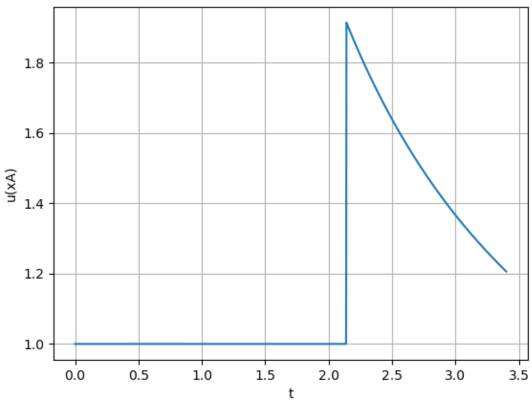


Figure 8: Configuration 2: time history of the solution at the probe location $x_A = 1.2$. The value stays at the background $u = 1$ until the leading shock reaches the probe, where it jumps to the state immediately behind the front; it then decays as the trailing rarefaction fan subsequently passes over x_A.

Numerical diffusion The upwind scheme is only first-order accurate in space, and this accuracy comes at a cost. A Taylor analysis of the discrete equation shows that the leading truncation error of first-order upwinding has the form of an artificial diffusion term, that is, a term proportional to the second spatial derivative of u, with a coefficient proportional to the mesh spacing,

$$\Gamma_{\text{num}} \sim \frac{|u|\ \Delta x}{2}. \tag{48}$$

This *numerical* (or false) diffusion is not present in the original inviscid equation; it is introduced by the discretization. Its principal effect is to smear discontinuities over several cells, so that a shock that is mathematically sharp is represented numerically by a steep but finite transition. Because the coefficient is proportional to the mesh spacing Δx, the smearing is reduced as the mesh is refined.

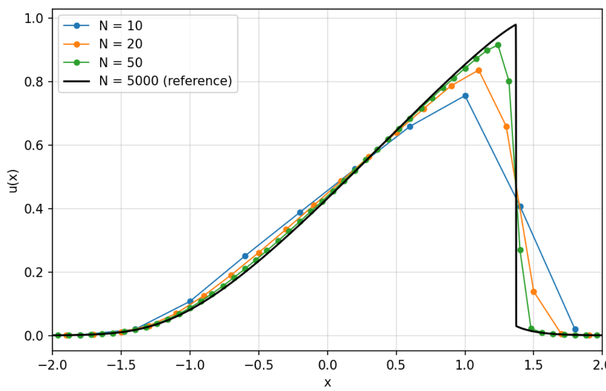


Figure 9: Effect of numerical diffusion on the solution of Configuration 1 for three mesh resolutions. As the number of control volumes increases, the artificial diffusion, proportional to the mesh spacing Δx, decreases: the leading shock front steepens towards its true vertical profile and the peak value is recovered more accurately.

This effect is illustrated in Figure 9, which compares the solution of Configuration 1 at $t = 1.5$ for several mesh resolutions against a fine-mesh reference. The computed cell values are shown as markers, so that the behaviour of the discrete solution itself—rather than of the interpolating line—is made visible. Two observations confirm that the discrepancy is due to numerical diffusion and not merely to the smaller number of points on a coarse mesh. First, away from the shock, where the solution is smooth, every resolution coincides with the reference. This follows from the form of the numerical diffusion, whose contribution to the equation,

$$\Gamma_{\text{num}} \frac{\partial^2 u}{\partial x^2}, \tag{49}$$

is proportional to the second spatial derivative of the solution. In the smooth regions this derivative is small and the error is negligible; at the shock it is large, so the error is confined to the region of steep gradient. Second, at the shock the coarse meshes both lower the peak value and spread the front over several cells, and as the mesh is refined the peak is progressively recovered and the front steepens towards the reference. Both observations identify the discrepancy as the numerical diffusion of the first-order upwind scheme: it is localized where the gradient is steep and it diminishes in proportion to the mesh spacing, confirming that it is an artifact of the discretization rather than of the graphical representation.

2.7 Verification against analytical solutions

The numerical results are now compared against exact solutions of the inviscid Burgers equation. No closed-form solution exists for arbitrary data, so the comparison is made in the regimes where an exact solution is available: each configuration is examined at a time chosen so that the analytical solution still holds.

Configuration 1. While the solution remains smooth, the equation is solved exactly by the method of characteristics: each value is transported at its own speed, giving the implicit solution

$$u(x,t) = u_0(x - u\,t)\,, \qquad u_0(\xi) = e^{-2\xi^2}. \tag{50}$$

This holds only until the characteristics first cross and a shock forms, at the breaking time

$$t_b = \frac{-1}{\min\limits_x u_0'(x)} \approx 0.824. \tag{51}$$

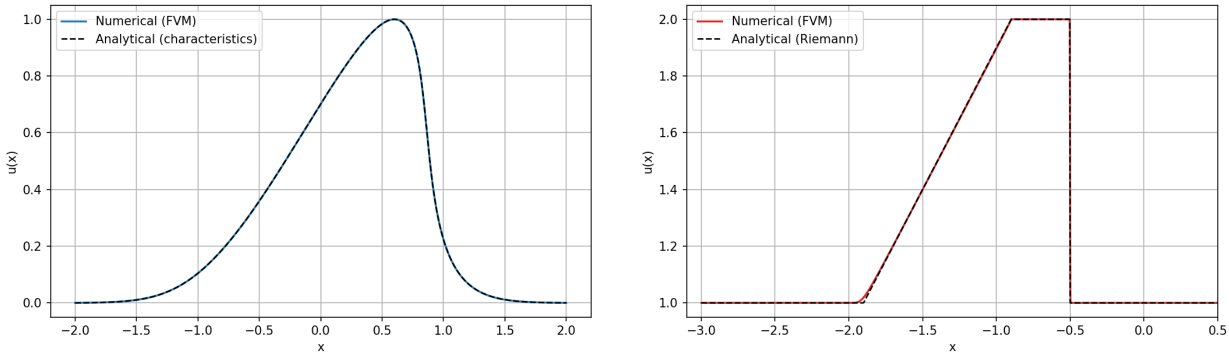
The comparison is therefore made at $t = 0.6 < t_b$, a time at which the field is still smooth and the analytical solution is exact. Figure 10a shows that the numerical profile coincides with the characteristic solution across the whole domain.

Configuration 2. The square-wave initial data is piecewise constant, so each edge of the block is an elementary Riemann problem. The leading edge is a shock travelling at the Rankine–Hugoniot speed $s = 3/2$, with position $x_{\text{shock}}(t) = -2 + \frac{3}{2}t$, while the trailing edge is a rarefaction fan,

$$u(x,t) = \begin{cases} 1, & x \le -2.9 + t, \\ \dfrac{x + 2.9}{t}, & -2.9 + t \le x \le -2.9 + 2t, \\ 2, & x \ge -2.9 + 2t. \end{cases} \tag{52}$$

This solution is valid only until the rarefaction overtakes the shock, which occurs at the interaction time $t_{\text{int}} = 1.8$; after that the two waves meet and the simple expressions no longer apply. The comparison is therefore made at $t = 1.0 < t_{\text{int}}$, while the shock and the fan are still independent. Figure 10b shows that the numerical solution reproduces the plateau, the rarefaction slope and the shock position.

In both cases the agreement in the smooth regions confirms that the scheme transports the solution correctly, while the discrepancy is confined to the discontinuities and is consistent with the numerical diffusion of the first-order upwind scheme.



(a) Configuration 1 at $t = 0.6$ (pre-shock).

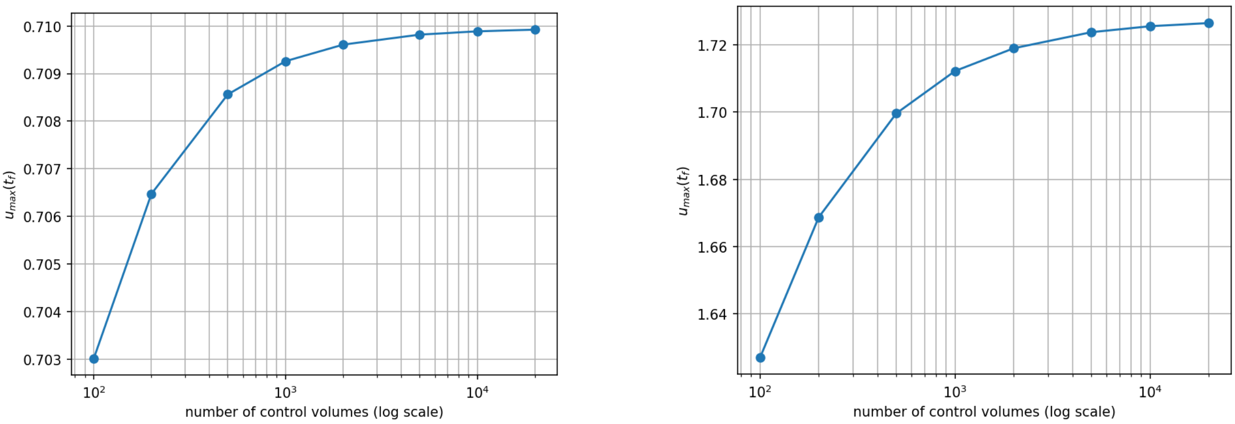
(b) Configuration 2 at $t = 1.0$ (pre-interaction).

Figure 10: Numerical solution (solid) versus analytical solution (dashed). Each comparison is made at a time at which the exact solution is still valid: before shock formation for Configuration 1, and before shock–rarefaction interaction for Configuration 2.

2.8 Mesh independence

To confirm that the working resolution resolves the solution adequately, a mesh independence study was carried out for both configurations. The peak value at the final time, $u_{\max}(t_f)$, was monitored as the number of control volumes was increased from $N = 100$ to $N = 20000$. This quantity is a sensitive indicator of numerical diffusion, since the first-order upwind smearing suppresses the peak most strongly on coarse meshes.

Figure 11 shows $u_{\max}(t_f)$ as a function of N on a logarithmic scale. In both configurations the peak value increases monotonically and converges to a single value, the curve flattening for $N \gtrsim 5000$. The monotone approach from below is the expected signature of first-order upwind diffusion, whose coefficient scales as $\Gamma_{\text{num}} \sim \frac{1}{2}|u|\,\Delta x$ and therefore vanishes linearly with the cell size; the most accurate peak value is recovered as $\Delta x \to 0$, as it was highlighted in Figure 9.



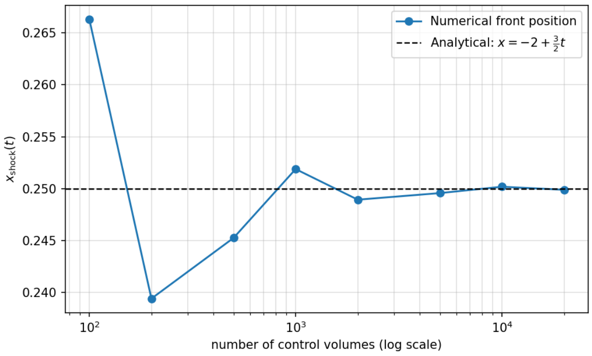
(a) Configuration 1.

(b) Configuration 2.

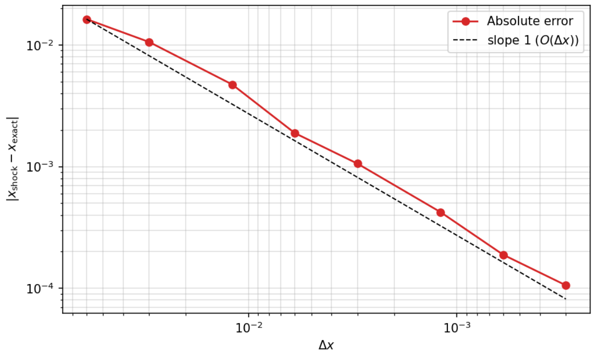
Figure 11: Mesh independence: peak value $u_{\max}(t_f)$ at the final time as a function of the number of control volumes N (logarithmic scale). In both configurations the peak increases monotonically and converges for $N \gtrsim 5000$.

A complementary check is obtained by tracking a geometric feature of the solution rather than its peak value. For Configuration 2 the leading shock has an exact position before the rarefaction reaches it: while $t < t_{\text{int}} = 1.8$ the front travels at the Rankine–Hugoniot speed $s = 3/2$, so its location is $x_{\text{shock}}(t) = -2 + \frac{3}{2}t$. The discrete front position, determined as the crossing of the mid-shock value, was measured at $t = 1.5$ and compared against this analytical reference for the same sequence of meshes.

Figure 12 shows the result. The numerical front converges monotonically to the exact position $x_{\text{shock}} = 0.25$ as the mesh is refined (Figure 12a); the scatter on the coarsest meshes reflects the cell-level ambiguity in locating the front. The absolute error (Figure 12b) decreases along a slope-one reference line on logarithmic axes, confirming that the shock position converges at first order, $O(\Delta x)$, in agreement with the formal accuracy of the first-order upwind scheme.



(a) Front position versus mesh, with the analytical reference.



(b) Absolute error versus Δx (log–log).

Figure 12: Configuration 2: convergence of the leading-shock position at $t = 1.5$ towards the exact value $x_{\text{shock}} = -2 + \frac{3}{2}t$. The front position converges to the analytical value (a), and the absolute error decreases at first order, $O(\Delta x)$ (b).

3 Conclusion

The inviscid Burgers equation was solved in one dimension by an explicit finite-volume method, using first-order upwind fluxes in conservative form and an adaptive time step at Courant number $C = 1$. Two configurations were considered: a smooth Gaussian pulse and a discontinuous square wave. Together they cover the two characteristic behaviours of a nonlinear hyperbolic equation, namely the spontaneous formation of a shock from smooth data and the transport of a discontinuity present from the start.

The numerical results reproduce the expected physics in both cases. The Gaussian pulse steepens on its forward face until a shock forms at the breaking time $t_b \approx 0.824$, after which the peak is convected downstream and its amplitude decays monotonically, in agreement with the convex combination principle. The square wave develops a leading shock travelling at the Rankine–Hugoniot speed $s = 3/2$ and a trailing rarefaction fan, with the peak amplitude held constant until the rarefaction erodes it.

The scheme was verified against exact solutions where these are available: the characteristic solution before shock formation for the Gaussian, and the Riemann solution before wave interaction for the square wave. In both cases the numerical profile agrees with the analytical one to within a small numerical error, located across the discontinuities. This residual error is the numerical diffusion intrinsic to the first-order upwind scheme, whose coefficient scales linearly with the mesh spacing. The mesh independence study confirmed this interpretation: the peak value converges monotonically and the shock position converges to its exact location at first order, $O(\Delta x)$, in agreement with the formal accuracy of the scheme.

The choice of the conservative flux form proved essential. Only this form reproduces the correct propagation speed of discontinuous solutions, since the non-conservative form is undefined across a shock. The first-order upwind scheme, however, introduces numerical diffusion that smears sharp fronts on coarse meshes; this is its principal limitation, and it is reduced only by refining the mesh. The method therefore captures shock formation, propagation, and rarefaction reliably and without oscillations, its only cost being a numerical diffusion that is well understood and controllable through mesh refinement.

4 Declaration of AI use

In accordance with the faculty guidelines on the use of artificial-intelligence tools in academic work, this section declares the use of such tools in the preparation of the present report.

A generative-AI assistant, Claude (Anthropic, model Opus 4.8) [4], was used as a technical aid during the development of the work. Its use was confined to a supporting role and did not replace the author's analysis, implementation, or conclusions. Specifically, the assistant was used for the following purposes:

– to clarify conceptual points of the nonlinear hyperbolic problem, including the meaning of characteristics, the derivation of the breaking time t_b, the Rankine–Hugoniot condition, and the admissibility of the Courant number $C = 1$ for the scalar Burgers equation;

– to discuss the treatment of the outflow boundary and the reason the fixed right-boundary value does not influence the solution under upwind differencing;

– to assist with the wording and structure of the report in English, and with the formatting of equations, tables and figures in LaTeX, including the TikZ diagrams of the characteristics, the shock-formation figure, and the outflow-boundary sketch;

– to produce, from the problem specification, an independent solution whose results were compared against those of the author's own code, as a cross-validation of the numerical results;

– to help set up the analytical comparisons (characteristic and Riemann solutions) and the mesh-independence and shock-position analyses.

Every suggestion produced by the assistant was independently verified before being incorporated. The numerical results reported here were obtained exclusively from the author's own Python code (`1D_convection.py`); the algebraic derivations were re-derived and checked by hand against the course notes; and the physical arguments were confirmed against the computed solution profiles, the evolution of the maximum value, and the analytical shock speeds. No text, equation, or result was accepted without this verification.

The artificial-intelligence tool is acknowledged solely as a technical aid and is not, in any sense, an author or co-author of this work. Full responsibility for the contents, interpretations, and conclusions of the report rests with the author.

Upon completion, the source code (`1D_convection.py`) and the present manuscript were submitted to a generative-AI assistant, Claude (Anthropic, model Opus 4.8), for an independent review of their internal consistency.

References

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